

Esben Jannik Bjerrum, Senior Cheminformatician at AbbVie (previously at AstraZeneca), co-developer of <https://github.com/MolecularAI/aizynthfinder>

AstraZeneca reaction data

- Esben said that AstraZeneca had its own reaction data generated by its medicinal chemists.
- Even these internal datasets were very noisy and often lacked good provenance.
- In many cases, it was difficult to trace back important experimental details such as how long a reaction had been run, the exact amounts used, which reagents were used, and on which equipment the experiment had been performed.
- He repeatedly referred to this missing information as the **context** of the chemist and the reaction.

Reaction templates and neural-network approach

- He described an approach based on approximately **160,000 reaction SMARTS templates**.
- A neural network was used to map a product to the reaction templates that could have produced it.
- Given a product, the model generated a probability distribution over the possible reaction templates, effectively estimating which transformations were most likely to produce that product.
- He also noted that **Pistachio is not the largest reaction dataset, but it is one of the cleanest**.

Dataset curation and machine learning performance

- According to Esben, a large part of his and his colleagues' time at AstraZeneca was spent curating, cleaning, and working with these datasets rather than developing the machine learning itself.
- Despite having access to proprietary AstraZeneca data, the performance of the machine learning algorithms was disappointing.

Limitations of public reaction data

- He believes the problem is even more difficult when working with public-domain reaction data.

- In his view, existing retrosynthesis software has probably already extracted close to the maximum useful information that can be obtained from the currently available public datasets.
- Therefore, further improvements based only on the same public-domain data are likely to be limited.

Need for high-quality robotic data generation

- Esben believes that meaningful improvements will require the generation of new, high-quality reaction data using robotic facilities.
- Robotic experiments could provide more reliable and systematically recorded reaction data, including the experimental context and provenance that are often missing from datasets generated manually by medicinal chemists.
- He sees this type of high-quality, well-contextualised data as necessary for improving the current machine learning algorithms.

Maciej Wójcikowski, Senior Cheminformatician at Merck,
developer of [Merck's Synthia retrosynthesis software](#)

Retrosynthesis data

- The main commercial reaction datasets discussed were **Pistachio**, **SPRESI**, and **Thieme**.
- These are commercial datasets. **Pistachio**, from **NextMove Software**, was mentioned as the most affordable option.
- A major limitation in retrosynthesis is **data availability and quality**.
- Public and open-source reaction databases are generally not sufficient for this problem.

Retrosynthesis confidence estimation

- We discussed confidence estimation for retrosynthesis predictions.
- This is not a fully solved problem, but confidence estimation has already been implemented in multiple open-source retrosynthesis tools.
- The AI Center may already be using some of these tools in its current calculations.

Focusing on specific scaffolds or chemotypes

- We discussed whether it would make sense to narrow the problem to particular chemotypes or scaffolds for which more data are available.
- However, this was not considered specifically a retrosynthesis problem. A particular scaffold can already be selected and a chemical library built around it.
- The broader question is whether there are regions of chemical space where additional AI or machine learning methods would be particularly beneficial.
- Retrosynthesis remains difficult, with the availability of suitable training data being one of the main limitations.

Docking, co-folding, and fragment docking

- We discussed combining retrosynthesis with fragment docking and what the potential novelty in docking would be.
- Standard fragment docking itself is straightforward: a docking program is used to dock fragments.
- Co-folding and flexible docking were considered related problems. When both the protein and ligand are treated as flexible, the problem becomes more similar to co-folding and offers more opportunities for machine-learning approaches.
- The main distinction is conventional docking against a static receptor structure, where the protein is kept fixed.

SYNTHIA

- **SYNTHIA**, developed by **Merck**, was mentioned as an important reference for retrosynthesis.
- **SynthiaOnline.com** provides useful materials, including publications, presentation slides, webinars, links, and other resources.

Roger Sayle, CEO & Founder of Nextmove Software Ltd,
Provider of the [Pistachio reaction dataset](#)

Retrosynthesis as a long-standing problem

- Roger said that retrosynthesis has been studied since the 1970s.
- There have been several projects over the years, but according to him, none of them managed to generate enough value to recover the money invested in them.

Data as the main limitation

- He agreed that retrosynthesis is fundamentally a data problem.
- If you do not have your own robotic facilities for generating clean and reliable chemical reaction data, there is limited value in working repeatedly with the same existing public datasets or even with Pistachio.
- Many software competitors use the same datasets to build retrosynthesis tools, so using the same data does not by itself create much differentiation.
- In practice, a large part of the work goes into curating reaction datasets and creating high-quality data rather than into the machine learning itself.

Monte Carlo approach

- Roger found the proposed Monte Carlo approach unusual and rather bizarre.
- He said he had not encountered retrosynthesis approaches based on random number generation and that more deterministic approaches are usually used.
- He therefore doubted that a Monte Carlo approach would work well and suggested that there is likely a reason why it is not commonly used.

Pistachio

- Roger said that Pistachio is a standard retrosynthesis dataset used by many groups.
- The academic licence costs approximately 2,000 British pounds:
<https://nextmovesoftware.com/PriceList.pdf>

Igor Tetko, Cheminformatics Group Leader at Helmholtz, Munich

Developer of NLP transformers for retrosynthesis

<https://www.nature.com/articles/s41467-020-19266-y>

Tetko's experience with Transformer retrosynthesis was that substantial gains could be obtained by improving training and data augmentation, and his group demonstrated this experimentally. But once those relatively straightforward modelling improvements have been exploited, the limiting factor increasingly becomes the reaction data themselves. Public USPTO-derived datasets are noisy, inconsistently curated and incomplete; some reactions are actually

impossible to reproduce under exact-match evaluation because reactants, reagents, solvents and conditions have been mixed together, and even the large USPTO-Full dataset contains erroneous records that need to be removed. Moreover, the standard USPTO-50k benchmark is too small and restricted to tell us whether a method will scale to realistic synthesis planning. Thus, simply building another Transformer or incrementally improving the architecture on the same public data is unlikely to produce a dramatic improvement in practical retrosynthesis. A more substantial advance would require cleaner, broader and more chemically diverse reaction data, better treatment of reaction conditions and alternatives, and evaluation at the level of complete synthesis routes rather than only single-step exact-match accuracy.

[Alexander Varkek, Prof. Cheminformatics, Strasbourg](#)

Developer of SynPlanner for retrosynthesis planning

<https://pubs.acs.org/jcis8/article-abstract/65/1/15/3686233/SynPlanner-An-End-to-End-Tool-for-Synthesis?redirectedFrom=fulltext>

Varnek was skeptical that an approach based on routinely applying DFT to candidate reaction steps could serve as a general retrosynthesis or reaction-prediction engine. Quantum chemistry is extremely valuable when the chemical system and the mechanistic question are sufficiently constrained, but automatic reaction prediction creates a combinatorial problem: one does not initially know which conformations, intermediates, transition states, elementary steps, catalysts, additives or competing pathways have to be calculated. Exhaustively exploring this space at DFT accuracy quickly becomes prohibitively expensive.

His collaboration with Satoshi Maeda's group at Hokkaido University illustrates precisely this problem. They use AFIR to search reaction pathways automatically and DFT as the high-quality reference, but already for comparatively modest systems a conventional DFT search is too expensive. Much of their current research is therefore devoted not to performing more DFT calculations, but to **avoiding most of them**—training specialised neural-network potentials on selected DFT calculations and then using those faster models to explore tens of thousands of possible reaction paths, with DFT retained for training, checking and refinement.

Their experience also shows why DFT should not be regarded as a universal solution for reaction prediction. Complex catalytic systems can contain huge conformational spaces and many competing pathways; mechanisms can change with substituents, catalysts, proton relays or reaction conditions; transition-metal systems can be particularly troublesome for cheaper approximate methods; and a small error in an activation barrier can strongly affect predicted kinetics and even direct an automated search into the wrong part of reaction space. For general retrosynthesis, the computational difficulty is therefore not simply calculating the energy of one

proposed transformation, but identifying which of the enormous number of possible transformations and pathways should be calculated in the first place.

On that basis, he regarded an architecture that attempted to use DFT broadly across the retrosynthetic search proposed in the ExPro abstract as computationally implausible. DFT could still have an important role as a **selective high-accuracy calculation** for carefully chosen difficult steps, mechanistic validation, transition-state refinement or generation of training data, but treating it as a general engine for exploring arbitrary retrosynthetic alternatives would face severe scaling problems.